

ENST2



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Normal Cones, Variational Inequalities, and Projections

A first course in nonsmooth contact and friction for geomechanics

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45 minute lecture

Why nonsmooth mechanics appears in geomechanics

Changing regimes

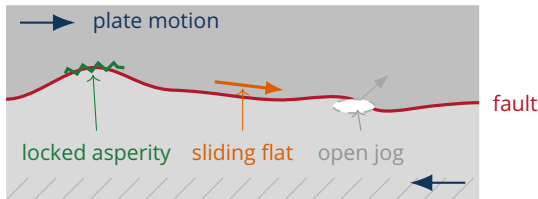
- A fault can be stuck or sliding.
- A rock joint can be open or closed.
- Crack faces can separate, touch, or slip.
- Granular particles can collide, stick, or slide.

The mathematical issue

- The law is not one smooth function.
- It contains inequalities.
- The reaction may be unknown inside a range.
- The active regime can change in time.

The key difficulty is not only geometry. It is **changing regime**.

One fault, three regimes at once



Real faults are never planar.
Roughness produces a mosaic of regimes coexisting on one fault:

- **locked asperities** — unknown stress;
- **sliding flats** — frictional limit μr_{ni} ;
- **open jogs** — no contact force.

One law, three regimes, one geometric idea.

A first coding exercise: what is $\text{Sign}(0)$?

Start from the scalar threshold-resistance model

$$\dot{x} + kx = f + r, \quad r \in -\text{Sign}(x).$$

Notebook prompt

At $x = 0$, the law gives the whole interval $\text{Sign}(0) = [-1, 1]$. Before showing the projection method, ask: what should the code return for r ?

We return to this in the spring-slider solver-logic slide.



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Option A: single-valued sign

$$r_{\text{code}}(x) = -\text{sign}(x), \quad \text{sign}(0) = 0.$$

```
r = -np.sign(x)
```

We return to this in the spring-slider solver-logic slide.



A first coding exercise: what is $\text{Sign}(0)$?

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Option A: single-valued sign

$$r_{\text{code}}(x) = -\text{sign}(x), \quad \text{sign}(0) = 0.$$

```
r = -np.sign(x)
```

Option B: regularized sign

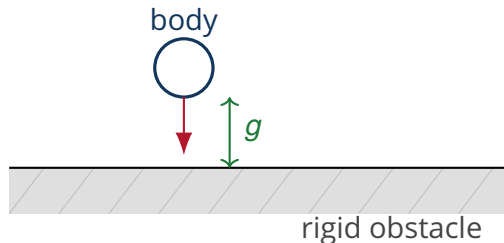
$$r_{\varepsilon}(x) = -\tanh(x/\varepsilon) \quad \text{or} \quad -\frac{x}{\sqrt{x^2 + \varepsilon^2}}.$$

```
r = -np.tanh(x/eps)
```

We return to this in the spring-slider solver-logic slide.



One running example: a point and a rigid wall



$$g \geq 0$$

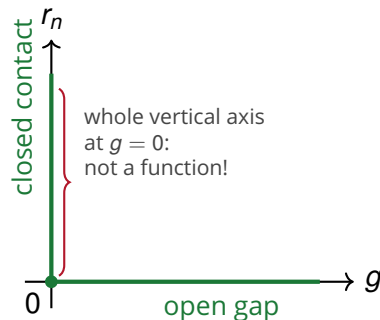
- $g > 0$: open gap.
- $g = 0$: contact.
- $g < 0$: impossible penetration.

We first understand this simple law, then add friction.

Signorini contact: a set-valued graph

$$g \geq 0, \quad r_n \geq 0, \quad g r_n = 0$$

- g is the normal gap.
- r_n is the compressive reaction.
- They cannot both be positive.

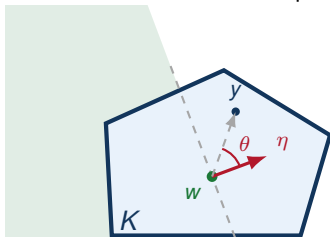


At $g = 0$, the reaction is set-valued: any $r_n \geq 0$ is admissible.

The normal cone: interior, edge, corner

$$N_K(w) = \{\eta : \langle \eta, y - w \rangle \leq 0 \forall y \in K\}.$$

Two equivalent tests for $\eta \in N_K(w)$: every $y - w$ makes angle $\theta \geq 90^\circ$ with η (*angle test*), or all of K lies in the half-plane behind η (*half-plane test*, shaded green).



× $\theta < 90^\circ$: this y violates $\langle \eta, y - w \rangle \leq 0$

Interior point. K surrounds w , so no perpendicular line through w keeps all of K behind it. Whatever direction we try for η , some $y \in K$ pokes out the wrong side.

$$N_K(w) = \{0\}$$

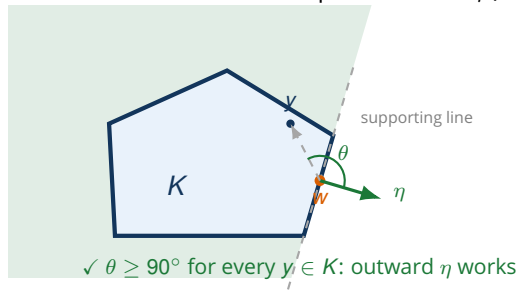
Only the trivial choice $\eta = 0$ survives



The normal cone: interior, edge, corner

$$N_K(w) = \{\eta : \langle \eta, y - w \rangle \leq 0 \forall y \in K\}.$$

Two equivalent tests for $\eta \in N_K(w)$: every $y - w$ makes angle $\theta \geq 90^\circ$ with η (*angle test*), or all of K lies in the half-plane behind η (*half-plane test*, shaded green).



Edge point. The supporting line at w (the edge itself) leaves K entirely on one side. The outward perpendicular η then makes $\langle \eta, y - w \rangle \leq 0$ for every $y \in K$. Tangential or inward η fails.

$$N_K(w) = \mathbb{R}_+ \eta_{\text{out}}$$

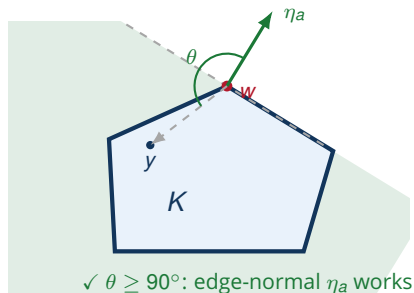
A single outward ray.



The normal cone: interior, edge, corner

$$N_K(w) = \{\eta : \langle \eta, y - w \rangle \leq 0 \ \forall y \in K\}.$$

Two equivalent tests for $\eta \in N_K(w)$: every $y - w$ makes angle $\theta \geq 90^\circ$ with η (*angle test*), or all of K lies in the half-plane behind η (*half-plane test*, shaded green).



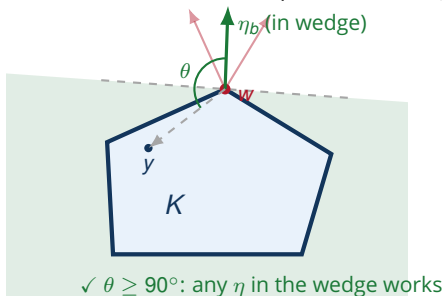
Corner — edge η_a . Take the outward normal of one of the two edges meeting at w . The perpendicular through w is that edge itself: K sits behind it. So this η_a is admissible. (Same argument applies to the other edge-normal η_b 's twin.)



The normal cone: interior, edge, corner

$$N_K(w) = \{\eta : \langle \eta, y - w \rangle \leq 0 \ \forall y \in K\}.$$

Two equivalent tests for $\eta \in N_K(w)$: every $y - w$ makes angle $\theta \geq 90^\circ$ with η (*angle test*), or all of K lies in the half-plane behind η (*half-plane test*, shaded green).



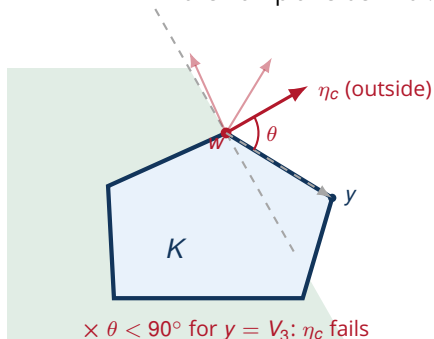
Corner — inside the wedge. Now try a η_b between the two edge-normals. The perpendicular tilts, but K is still entirely behind it. Any η in the wedge works.



The normal cone: interior, edge, corner

$$N_K(w) = \{\eta : \langle \eta, y - w \rangle \leq 0 \ \forall y \in K\}.$$

Two equivalent tests for $\eta \in N_K(w)$: every $y - w$ makes angle $\theta \geq 90^\circ$ with η (*angle test*), or all of K lies in the half-plane behind η (*half-plane test*, shaded green).



Corner — outside the wedge. Push η_c past the boundary. The perpendicular line cuts *into* K , and some $y \in K$ pokes out: fails.

$$N_K(w) = \text{wedge between edge-normals}$$



The normal cone captures Signorini

The compact geometric form is

$$g \in \mathbb{R}_+, \quad -r_n \in N_{\mathbb{R}_+}(g).$$

Open gap: $g > 0$

$$N_{\mathbb{R}_+}(g) = \{0\}.$$

Therefore

$$-r_n = 0 \quad \Rightarrow \quad r_n = 0.$$

Open gap, no reaction.

Closed contact: $g = 0$

$$N_{\mathbb{R}_+}(0) = (-\infty, 0].$$

Therefore

$$-r_n \leq 0 \quad \Rightarrow \quad r_n \geq 0.$$

Closed contact, compressive reaction.

$$g \geq 0, \quad r_n \geq 0, \quad gr_n = 0.$$

One set-valued inclusion contains all three Signorini clauses.

Projection identity: closest point $\Leftrightarrow$ normal cone

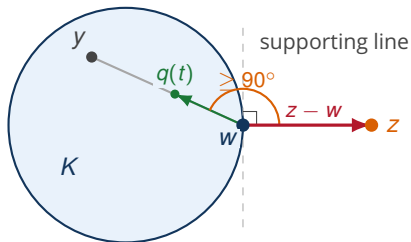
For a closed convex K , let $w = \text{proj}_K(z)$ be the closest point in K to an arbitrary z . The residual is $z - w$.

$$w = \text{proj}_K(z) \iff z - w \in N_K(w).$$

For Signorini: $K = \mathbb{R}_+$, $w = g$ (gap), residual $z - w = -\rho r_n$ (scaled reaction).

Projection and normal-cone inclusions are the same statement.

Projection geometrically: the residual is normal



Why the identity is true, geometrically.

$$w = \text{proj}_K(z) = \arg \min_{q \in K} \|q - z\|^2.$$

For any $y \in K$, the segment $q(t) = w + t(y - w)$, $t \in [0, 1]$, stays in K (convexity). Since w is the closest point, $\|q(t) - z\|^2$ is minimised at $t = 0$, so its derivative there is non-negative:

$$-2 \langle z - w, y - w \rangle \geq 0 \Rightarrow \langle z - w, y - w \rangle \leq 0.$$

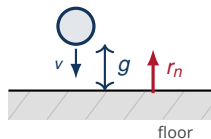
The angle between residual and any $y - w$ is $\geq 90^\circ$ — exactly $z - w \in N_K(w)$.

Coming up: apply this with $z - w = -\rho r_n$ and $w = g$ — Signorini becomes one projection equation

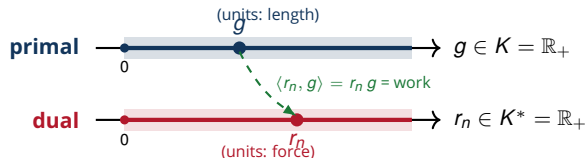


Primal and dual: kinematic vs kinetic

Kinematics (motion: g, v) and **kinetics** (forces: r_n) live in *different* spaces with different units. They are coupled through an energy bracket: $r_n g$ has units of *work*; in dynamic problems $r_n v$ has units of *power*.



physical setup



Same shape ($\mathbb{R}_+$), different spaces and units.

Complementarity: $g r_n = 0$. Either the body is off the floor ($g > 0, r_n = 0$) or in contact and silent or pushed ($g = 0, r_n \geq 0$). Bracket vanishes either way.

The pairing is symmetric in g and r_n : we can solve for either one.

Signorini as a fixed-point equation

Swap of roles. The complementarity $g \geq 0, r_n \geq 0, gr_n = 0$ is symmetric in (g, r_n) . So we may equally well treat the dual variable r_n as the unknown:

$$r_n \in \mathbb{R}_+, \quad -g \in N_{\mathbb{R}_+}(r_n).$$

Projection identity. Apply $w = \text{proj}_K(z) \Leftrightarrow z - w \in N_K(w)$ with $w = r_n, z = r_n - \rho g$ ($\rho > 0$):

$$r_n = \text{proj}_{\mathbb{R}_+}(r_n - \rho g) = \max(r_n - \rho g, 0).$$

One equation, both regimes; the right-hand side is just max. Ready for a solver.

Bouncing ball: solver logic

$$W = \frac{1}{m}, \quad v^{\text{free}} = v^n - gh.$$

Substitute $v^{n+1} = v^{\text{free}} + \frac{r^{n+1}}{m}$:

$$r^{n+1} = \text{proj}_{\mathbb{R}_+} \left(r^{n+1} - \rho \left[v^{\text{free}} + \frac{r^{n+1}}{m} \right] \right).$$

Pick $\rho = m$: the r^{n+1} inside vanishes,

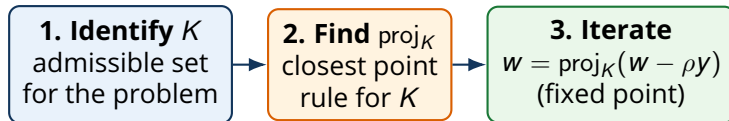
$$r^{n+1} = \text{proj}_{\mathbb{R}_+} (-m v^{\text{free}}).$$

Affine argument $\Rightarrow$ a clever ρ kills the implicit r^{n+1} .



Recipe: identify K , find proj_K , then iterate

The fixed-point form $w = \text{proj}_K(w - \rho y)$ is a recipe with three ingredients. The first two are problem-specific; the third is mechanical.



Read on Signorini.

- K : admissible reactions are non-negative, $K = \mathbb{R}_+$.
- proj_K : closest point in $\mathbb{R}_+$ is $\text{proj}_{\mathbb{R}_+}(z) = \max(z, 0)$.
- **Iterate**: apply with $w = r_n$, $y = g$. Equation $r_n = \max(r_n - \rho g, 0)$ already holds at the solution; iteration is one step in 1D.

Same three ingredients return for friction; only K changes.

From gap to velocity: the formulation we'll use henceforth

For dynamic problems — and for friction, which depends on slip rate — we work at the *velocity level*. Differentiate the constraint, replace gap with normal velocity:

Position level (so far)

$$g \geq 0, \quad r_n \geq 0, \quad g r_n = 0.$$

Velocity level (henceforth, at $g = 0$)

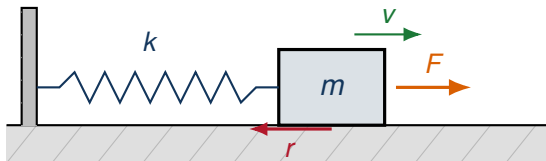
$$v_n \geq 0, \quad r_n \geq 0, \quad v_n r_n = 0.$$

Same shape, same recipe — velocity in the primal slot:

$$r_n = \text{proj}_{\mathbb{R}_+}(r_n - \rho v_n).$$

From state to rate: the framework is unchanged.

Running example for friction: a spring-slider



A block on a rough floor, pulled by a spring or external force F .

- As F grows, the friction reaction r grows in step — the block stays put.
- Once $|r|$ reaches the threshold μN , slip begins.
- During slip, r stays at $\pm\mu N$ and opposes v .

Same template, new admissible set: $K = [-\mu N, \mu N]$.

Spring-slider: solver logic

$$W = \frac{h}{1+hk}, \quad v^{\text{free}} = \frac{x^n + hf}{1+hk}.$$

Substitute $x^{n+1} = v^{\text{free}} + \frac{h}{1+hk} r^{n+1}$:

$$r^{n+1} = \text{proj}_{[-1,1]} \left(r^{n+1} - \rho \left[v^{\text{free}} + \frac{h r^{n+1}}{1+hk} \right] \right).$$

Pick $\rho = \frac{1+hk}{h}$: the r^{n+1} inside vanishes,

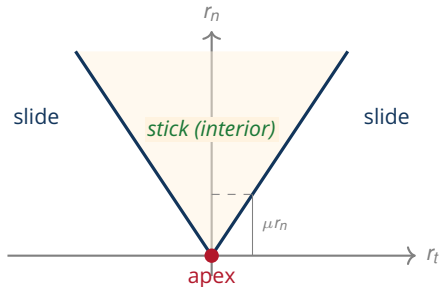
$$r^{n+1} = \text{proj}_{[-1,1]} \left(-f - \frac{x^n}{h} \right).$$

Same trick: affine argument $\Rightarrow$ a clever ρ kills the implicit r^{n+1} .



From a friction interval to the Coulomb cone

Lift to 2D: the normal load r_n is now part of the unknown, and r_t may point in any tangential direction. The admissible reactions form a cone in (r_n, r_t) -space.



$$K_\mu = \{(r_n, r_t) : r_n \geq 0, \|r_t\| \leq \mu r_n\}.$$

Slope μ , apex at the origin. Stick inside, slide on the boundary.

Friction + contact = a second-order cone.

Friction plus normal contact: regimes from the cone

One local reaction $r = (r_n, r_t)$ must satisfy

$$K_\mu = \{(r_n, r_t) : r_n \geq 0, \|r_t\| \leq \mu r_n\}.$$

Take-off



$$r = 0, \quad v_n \geq 0.$$

No contact reaction.

Sticking



$$v = 0, \quad \|r_t\| \leq \mu r_n.$$

The reaction may be interior or on the boundary.

Sliding



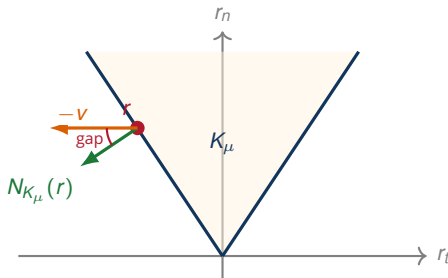
$$v_n = 0, \quad v_t \neq 0,$$

$$\|r_t\| = \mu r_n.$$

Boundary plus nonzero slip velocity.

Why ordinary normality points the wrong way

The naive guess. Carry the Signorini template across: $-v \in N_{K_\mu}(r)$. We will see this fails geometrically during sliding.



During steady sliding $v_n = 0$, $v_t \neq 0$, so $-v$ is purely *horizontal*. But on a smooth boundary point of K_μ , $N_{K_\mu}(r)$ is an *oblique* ray of slope μ :

$$-v \notin N_{K_\mu}(r) \text{ during sliding.}$$

The naive law misses the normal component $\mu\|v_t\|$ that the cone's geometry demands.

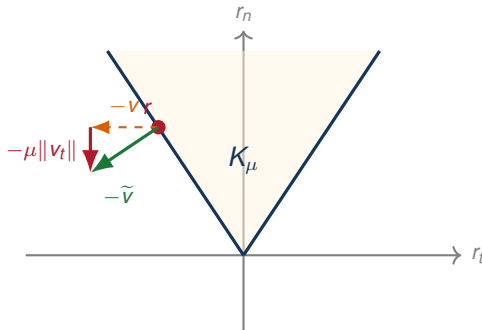
The directions don't match: a fix is needed.

De Saxcé shift: building orthogonality geometrically

The fix. A horizontal slip component $\|v_t\|$ needs a downward normal component $\mu\|v_t\|$. Set

$$\tilde{v} = (v_n + \mu\|v_t\|, v_t),$$

so $-\tilde{v}$ has the missing oblique direction added back.



After the shift, $-\tilde{v}$ lies along $N_{K_\mu}(r)$:

$$-\tilde{v} \in N_{K_\mu}(r).$$

Testing the normal-cone inequality with 0 and $2r$ (for $r \neq 0$) forces

$$\tilde{v} \cdot r = 0.$$

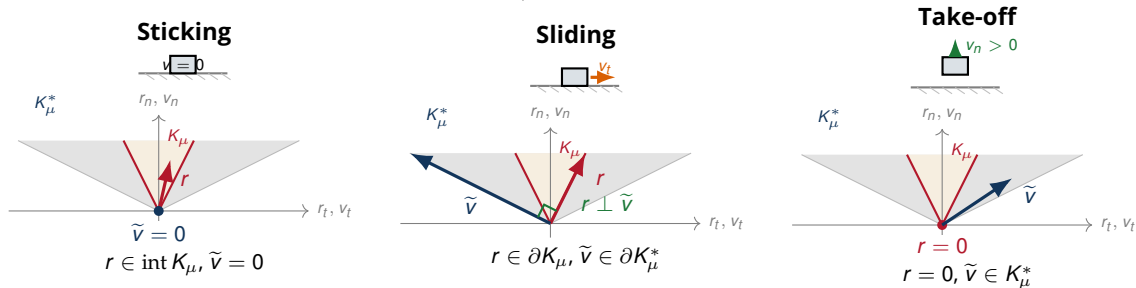
The shift creates orthogonality.

Three regimes from cone complementarity

The De Saxcé law $-\tilde{v} \in N_{K_\mu}(r)$ is equivalent to the boxed *cone-complementarity* statement

$$\boxed{r \in K_\mu, \quad \tilde{v} \in K_\mu^*, \quad r \cdot \tilde{v} = 0,} \quad K_\mu^* = \{w : \mu \|w_t\| \leq w_n\} \text{ (dual cone, slope } 1/\mu\text{)}.$$

The three Coulomb regimes drop out as the three ways this can hold: r in $\text{int } K_\mu$ with $\tilde{v} = 0$; both on respective boundaries; or $r = 0$ with $\tilde{v}$ free in K_μ^* .

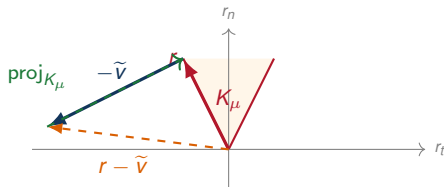


One algebraic statement, three physical regimes.

Projection geometrically: $r = \text{proj}_{K_\mu}(r - \tilde{v})$

Same trick as Signorini. The projection optimality $w = \text{proj}_K(z) \Leftrightarrow z - w \in N_K(w)$ turns the law $-\tilde{v} \in N_{K_\mu}(r)$ into

$$r = \text{proj}_{K_\mu}(r - \tilde{v}).$$



$-\tilde{v}$ points along the outward normal at r , so the trial point $r - \tilde{v}$ lies on the perpendicular through r to ∂K_μ .

Projecting back onto K_μ recovers r exactly.

$\rho > 0$ may be inserted:

$r = \text{proj}_{K_\mu}(r - \rho \tilde{v})$, since N_{K_μ} is a cone.

Coulomb 2D: solver logic

W matrix; the law uses the De Saxcé shift $\tilde{v} = (v_n + \mu \|v_t\|, v_t)$.

Recipe:

$$r^{n+1} = \text{proj}_{K_\mu}(r^{n+1} - \rho \tilde{v}),$$

where $v^{n+1} = v^{\text{free}} + W r^{n+1}$, then $\tilde{v}$ from v^{n+1} .

No ρ kills the dependence: $\tilde{v}$ involves $\|v_t\|$, which is *non-affine* in r^{n+1} .

must iterate the recipe.

Non-affine argument $\Rightarrow$ iterate.



Recipe for nonsmooth contact mechanics

The whole lecture compressed into one template: **identify** K , **find** proj_K , **iterate the fixed point**. Three problems, three specializations.

	Admissible set K	Projection proj_K	Fixed-point equation
Signorini (point + wall)	$K = \mathbb{R}_+$ non-negative reactions	$\max(z, 0)$ clamp at zero	$r_n = \text{proj}_{\mathbb{R}_+}(r_n - \rho g)$
Friction (1D) (spring-slider)	$K = [-\mu N, \mu N]$ bounded interval	$\max(-\mu N, \min(\mu N, z))$ clip to interval	$r = \text{proj}_{[-\mu N, \mu N]}(r - \rho v)$
Coulomb (2D) (planar contact)	$K_\mu = \{r_n \geq 0, \ r_t\ \leq \mu r_n\}$ second-order cone	shrink-radial onto K_μ closed-form	$r = \text{proj}_{K_\mu}(r - \rho \tilde{v})$ (De Saxcé shift)

What changes: the geometry of K — a half-line, an interval, a 2D cone. **What stays the same:** the set-valued law $-y \in N_K(w)$ and the fixed-point form $w = \text{proj}_K(w - \rho y)$.

Three problems, one template.

Supplemental S0a — The contact reaction as a measure

Measure, in plain words. A way to count “how much reaction” over any time interval, including a single instant. Bounded forces deposit *nothing* at a point; Diracs deposit their full weight.

Decomposition: the reaction measure splits into two pieces.

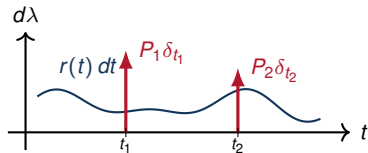
$$d\lambda = \underbrace{r(t) dt}_{\text{smooth force}} + \underbrace{\sum_k P_k \delta_{t_k}}_{\text{impact percussions}} .$$

Equation of motion: $M d\dot{q} = F dt + H^\top d\lambda$.

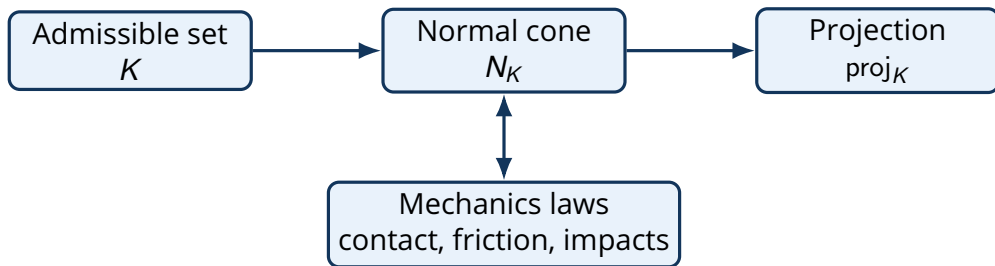
Same cone law on each piece:

- Smooth interval: $r(t) \in K_\mu$, $-\Phi(v(t)) \in N_{K_\mu}(r(t))$.
- Impact t_k : $P_k \in K_\mu$, $-\Phi(v^+) \in N_{K_\mu}(P_k)$ on the post-impact velocity (**Moreau**: this enforces dissipation).

Same cone, two time scales — one equation.



Supplemental S0b — The whole lecture in one diagram



The reaction is the geometry of the constraint.

Supplemental S0c — Cone complementarity and the dual cone

The *dual cone* is the set of vectors making a non-obtuse angle with every admissible reaction:

$$K_{\mu}^* = \{w : w \cdot s \geq 0 \forall s \in K_{\mu}\} = \{(w_n, w_t) : \mu \|w_t\| \leq w_n\}.$$

For the De Saxcé shifted velocity, $\tilde{v} \in K_{\mu}^*$ (see S2). Together with $r \in K_{\mu}$, the orthogonality from S1 gives *cone complementarity*:

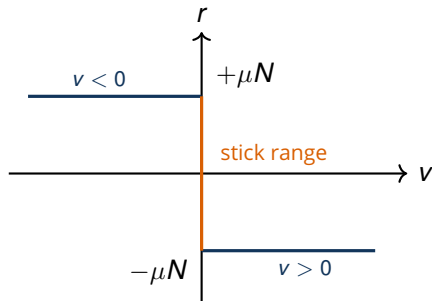
$$\boxed{r \in K_{\mu}, \quad \tilde{v} \in K_{\mu}^*, \quad \tilde{v} \cdot r = 0} \iff -\tilde{v} \in N_{K_{\mu}}(r).$$

Then the same projection identity as Signorini gives

$$\boxed{r = \text{proj}_{K_{\mu}}(r - \rho \tilde{v})}, \quad \rho > 0.$$

The 2D friction law has the same fixed-point shape; the shift was the missing geometry.

Supplemental S0d — Friction in 1D: graph and cone form



Same template as Signorini: a reaction confined to an admissible set, with the negative velocity in its normal cone.

$$r \in K = [-\mu N, \mu N], \quad -v \in N_K(r).$$

- $|r| < \mu N$: $N_K(r) = \{0\}$, so $v = 0$ — **stick**.
- $r = +\mu N$: $-v \in [0, \infty)$, so $v \leq 0$ — stick or slide leftward.
- $r = -\mu N$: symmetric; stick or slide rightward.

Same template as Signorini, interval instead of $\mathbb{R}_+$: $r = \text{proj}_{[-\mu N, \mu N]}(r - \rho v)$.

Supplemental S0e — Why we need a measure: impacts

The pointwise contact law $r(t) \in K_\mu$, $-\Phi(v(t)) \in N_{K_\mu}(r(t))$ works for sustained contact and stick-slip — but **not** for impacts.

An **impact** (a ball hitting the floor) produces an instantaneous velocity jump. By Newton's second law, that jump is the impulse:

$$M(\dot{q}^+ - \dot{q}^-) = \int_{t_k - \varepsilon}^{t_k + \varepsilon} r(t) dt.$$

The problem. If r is a bounded function ($\|r(t)\| \leq C$), then for any ε

$$\left\| \int_{t_k - \varepsilon}^{t_k + \varepsilon} r(t) dt \right\| \leq 2C\varepsilon \xrightarrow{\varepsilon \rightarrow 0} 0,$$

which contradicts a finite jump on the left.

The fix. r must be a *Dirac measure* concentrated at t_k — not a function, but a distribution.

Same contact law. New time object.

Supplemental S1 — Orthogonality from the cone law

At one contact point, the unknown is the reaction

$$r = (r_n, r_t), \quad K_\mu = \{r : r_n \geq 0, \|r_t\| \leq \mu r_n\}.$$

De Saxcé introduces $\tilde{v} = \Phi(v) = (v_n + \mu \|v_t\|, v_t)$. The contact law is

$$r \in K_\mu, \quad -\tilde{v} \in N_{K_\mu}(r).$$

By the normal-cone inequality, $\tilde{v} \cdot (s - r) \geq 0$ for every $s \in K_\mu$. Assume first $r \neq 0$. Because K_μ is a cone, both 0 and $2r$ are admissible tests:

$$s = 0 : \tilde{v} \cdot r \leq 0; \quad s = 2r : \tilde{v} \cdot r \geq 0.$$

Hence

$$\tilde{v} \cdot r = 0.$$

For $r = 0$, the orthogonality holds trivially. **Reaction and shifted velocity are orthogonal — cone complementarity.**

Supplemental S2 — $\tilde{v} \in K_\mu^*$ directly

Recall the dual cone, by definition:

$$K_\mu^* = \{y = (y_n, y_t) : y_n \geq \mu \|y_t\|\}.$$

So showing $\tilde{v} \in K_\mu^*$ means showing $\tilde{v}_n \geq \mu \|\tilde{v}_t\|$. Plug in $\tilde{v} = (v_n + \mu \|v_t\|, v_t)$:

$$\tilde{v}_n = v_n + \mu \|v_t\| \geq \mu \|v_t\| = \mu \|\tilde{v}_t\|,$$

where the inequality uses the kinematic no-penetration condition $v_n \geq 0$, independent of the contact law. Done:

$$\tilde{v} \in K_\mu^*.$$

Combined with $r \in K_\mu$ (admissibility) and $\tilde{v} \cdot r = 0$ (S1), this is the full cone-complementarity statement

$$r \in K_\mu, \quad \tilde{v} \in K_\mu^*, \quad \tilde{v} \cdot r = 0.$$

Coulomb law $\Rightarrow$ cone complementarity. (S3 does the converse via case analysis.)

Supplemental S3 — Regime extraction

From cone complementarity, with $\tilde{v} = (v_n + \mu\|v_t\|, v_t)$:

The dual-cone condition $\tilde{v} \in K_\mu^*$ gives $v_n + \mu\|v_t\| \geq \mu\|v_t\| \geq 0$.

Orthogonality gives $0 = (v_n + \mu\|v_t\|)r_n + v_t \cdot r_t$. Since $\|r_t\| \leq \mu r_n$, Cauchy-Schwarz gives

$$v_t \cdot r_t \geq -\|v_t\|\|r_t\| \geq -\mu\|v_t\|r_n.$$

Therefore

$$\tilde{v} \cdot r = v_n r_n + \mu\|v_t\|r_n + v_t \cdot r_t \geq v_n r_n.$$

Since $\tilde{v} \cdot r = 0$ and $v_n r_n \geq 0$, we get $v_n r_n = 0$: contact cannot simultaneously separate and compress. Without orthogonality, $r \in K_\mu$ only bounds magnitude; it does not switch opening contacts off.

Regimes

- $r = 0$: take-off (inactive contact).
- $v = 0$: sticking, possibly on ∂K_μ .
- $v_t \neq 0$: Cauchy-Schwarz is sharp $\Rightarrow v_n = 0, \|r_t\| = \mu r_n, r_t \propto -v_t$.

Supplemental S4 — Backward-Euler algebra for the spring-slider

The spring-slider solver-logic slide claims that backward Euler applied to $\dot{x} + kx = f + r$ with $r \in -\text{Sign}(x)$ collapses to a single explicit projection. Here is the derivation.

The implicit time step is $x^{n+1} = (x^n + h(f + r^{n+1}))/ (1 + hk)$. The cone form is $-x \in N_{[-1,1]}(r)$ (Fenchel duality with $-r \in \text{Sign}(x)$). The projection identity gives, for any $\rho > 0$,

$$r^{n+1} = \text{proj}_{[-1,1]}(r^{n+1} - \rho x^{n+1}).$$

Substitute the dynamics into the right-hand side:

$$r^{n+1} - \rho x^{n+1} = r^{n+1} \left[1 - \frac{\rho h}{1 + hk} \right] - \frac{\rho(x^n + hf)}{1 + hk}.$$

Choose $\rho_h = (1 + hk)/h$. The bracket vanishes; the dependence on the unknown r^{n+1} on the right disappears, and what remains is just $-f - x^n/h$. Therefore

$$r^{n+1} = \text{proj}_{[-1,1]} \left(-f - \frac{x^n}{h} \right), \quad x^{n+1} = \frac{x^n + h(f + r^{n+1})}{1 + hk}.$$

Stick verification. If $|f + x^n/h| \leq 1$, the projection is the identity, so $r^{n+1} = -f - x^n/h$. Then $x^{n+1} = (x^n + h(f + (-f - x^n/h)))/ (1 + hk) = (x^n - x^n)/ (1 + hk) = 0$ exactly. The set-valued $-\text{Sign}(0) = [-1, 1]$ is consumed by the projection, with no smoothing parameter and no residual drift.

Supplemental S5 — Further reading

- **Normal cones, projections, optimality.**
G. Chierchia, *Optimization and Algorithms*, Ch. 5.
<https://perso.esiee.fr/~chierchg/optimization/>
- **Nonsmooth contact dynamics.**
V. Acary, B. Brogliato, *Numerical Methods for Nonsmooth Dynamical Systems*, Springer LNACM 35, 2008.
- **De Saxcé bipotential.**
G. de Saxcé, Z.-Q. Feng, "New inequality and functional for contact with friction," *Mathl. Comput. Modelling* 28(4–8):225–245, 1998.
- **Moreau's measure formulation.**
J. J. Moreau, "Unilateral contact and dry friction in finite freedom dynamics," in *Nonsmooth Mechanics and Applications*, CISM 302, Springer, 1988.